

Human and Animal Brains

The human brain is one of the defining features of our species. Comparing the human brain with the brains of other animals reveals connections between brain size and intelligence and between an animal's brain anatomy and the way it lives.

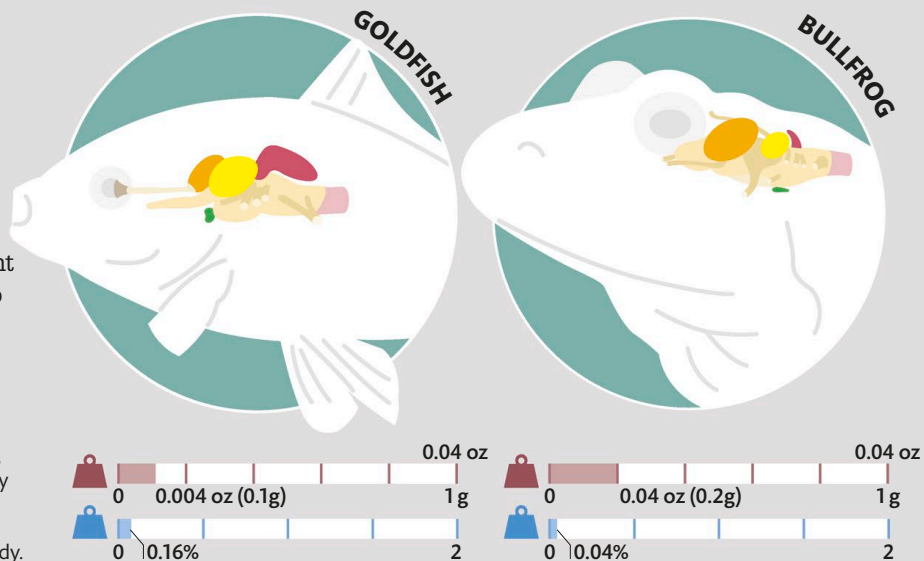
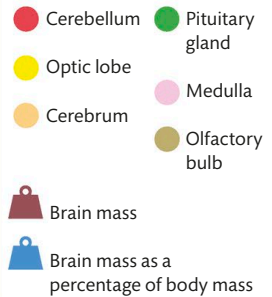
Brain sizes

The size of a brain indicates its total processing power. For example, a honeybee's tiny brain contains 1 million neurons, a Nile crocodile's has 80 million, while a human brain has around 80–90 billion neurons. The link with intelligence is clear. However, with larger animals, it is important to compare brain and body size to give a more nuanced indication of cognitive power.

Sizing up

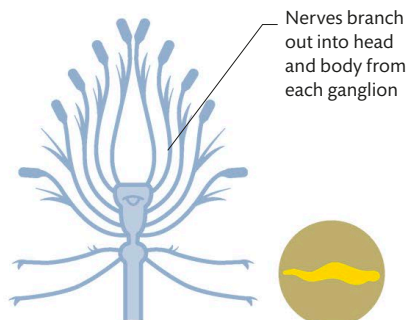
There are two ways to compare brain sizes, by total weight and as a percentage of body weight. The largest brain, at 17 lb (7.8 kg), belongs to the sperm whale, but that is a minute fraction of its 44-ton (45-tonne) body.

KEY



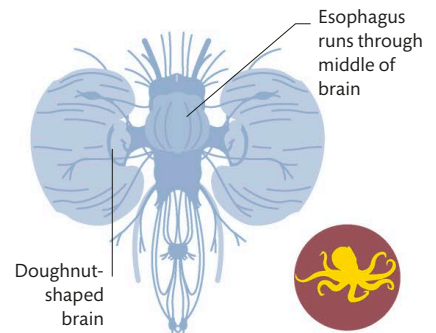
Brain shapes

All brains are located in the head, in close proximity to the primary sense organs. However, it would be a mistake to visualize animal brains as rudimentary variations, in size and structure, of the human brain. All vertebrate brains follow the same development plan, but anatomies vary widely to match different sensory and behavioral needs. More variety can be seen in the brains of invertebrates, which account for 95 percent of all animals.



Leech

The 10,000 cells in a leech's nervous system are arranged in chains of cell clusters called ganglia. The brain is a big ganglia, with 350 neurons, located at the front of the body.



Octopus

An octopus's brain contains 500 million neurons. Only a third are located in the head; the rest are in the arms and skin, where they are devoted to sensory and motor controls.